Design and Development of Owl-shaped Pet Companion Robots

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Abstract —**In the AIE1902 AI Toys program, we are developing an owl-shaped pet robot capable of interacting with humans and the environment—for example, flapping its wings when patted on the head and rotating its head when detecting a loud noise. In addition, the robot exhibits behaviors similar to those of a real owl, such as autonomously flapping its wings or chirping even without external stimulation. We will use Solidworks for shell modeling and then 3D print the shell. For coding, we will use the Arduino IDE and implement the expected functions using the ESP32 Dev Module. The robot is designed to be a cute and natural-looking pet, providing companionship and joy to its owner.**

Keywords— **Owl-shaped Robots, Interaction，Robot design, Design evaluation**

I. INTRODUCTION

A. Research Background - The social demand for low-burden emotional companionship

With the continuous decline in the global birth rate and the accelerating trend of population aging， more and more individuals are facing the dilemma of insufficient companionship and lack of daily care[1]. In modern society, the rapid pace of life， intensified work pressure[2], and the prevalence of living alone have further amplified people’s psychological needs for emotional sustenance and interactive companionship. Against this social backdrop， Raising pets has become one of the mainstream ways for people to relieve loneliness, ease life pressure, and obtain emotional comfort[3]. Nevertheless, traditional pet keeping is often associated with considerable demands in terms of time, financial expenditure, physical energy, living conditions, and health management, making long-term pet care difficult for many people to sustain[4]. In response to these constraints, care-free pet companion robots have emerged as a significant area of research, offering a low-maintenance and highly adaptable alternative for emotional companionship[5]-[7].Such robots are designed to provide accessible, consistent, and burden-free interaction, particularly for those unable to keep live pets due to personal or environmental constraints.

B. Problem Identification - Providing sufficient companionship for its owner As a small pet

Emotional companionship provied by pets plays an irreplaceable role in human psychological regulation[8][9]. On the one hand, the intuitive interaction, physical touch, and non-verbal feedback[10] of pets can effectively reduce people’s anxiety and depression, enhance the sense of security and presence, and meet the basic psychological needs of intimacy. On the other hand, the limitations associated with traditional pet keeping[11] have become increasingly evident in modern society. For an example, busy working hours[12] make it impossible to accompany pets for a long time; living space restrictions[13] are not suitable for raising large or active animals; physical discomfort, pet allergies[14], and travel arrangements[15] also hinder people from obtaining stable pet companionship. Existing companion robots still exhibit several limitations, including simple interaction modes[16], insufficient realism, limited response capability[17], and narrow applicability across user groups. As a result, they remain unable to fully simulate the natural, diverse, and real-time emotional feedback provided by real pets. Therefore, it is of great practical significance to develop a companion robot that is realistic and low-burden and can provide multi-modal interactive companionship [18]-[20].

C. Objectives

This project aims to develop an owl-shaped pet companion robot that could fulfill this purpose.

Specifically, the objectives include:

1)Achieving a lifelike owl appearance to enhance user affinity and psychological acceptance.
2)Implementing multi-modal interactive capabilities (e.g., sound, movement, touch response) to simulate natural pet-like feedback.
3)Ensuring low maintenance and adaptability to various living conditions and user needs.
4)Validating the robot’s effectiveness in providing companionship through user testing.

The robot is designed to closely resemble a true owl in appearance and to perform interactive functions that support sustained human-robot emotional engagement. Through the integration of realistic appearance design and multi-modal interactive functions, the proposed owl companion robot aims to offer users a sustainable, emotionally engaging, and low-burden alternative to live pet companionship. The robot developed, built, and tested was intended to have an appearance as a ture owl and perform interactive functions, ultimately fulfilling the need for companionship. Unlike existing companion robots with limited interactivity, this owl-shaped robot emphasizes real-time, multi-modal emotional feedback and adaptive responses, thereby bridging the gap between physical pet ownership and robotic companionship.

The following figure will present it.

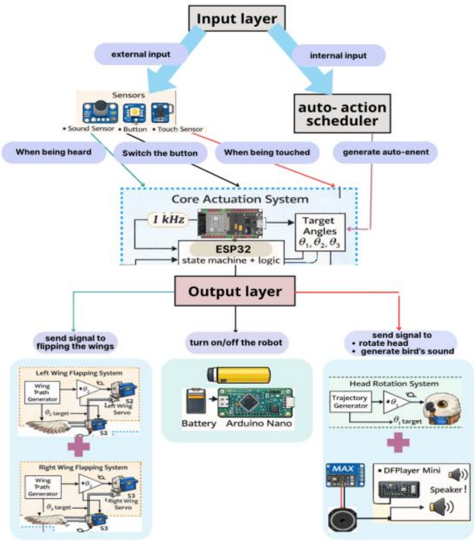


Fig. 1. The function diagram.

II. LITERATURE REVIEW

A literature review was conducted in several areas to ensure robot design and experimentation practices. Previous studies on pet-robot interaction in the field of robotics were examined to aiming to innovate and learn from prior work. Furthermore, the study references behavioral research on the appearance and behavior of owls to provide a basis for the robot's motion design and its 3D modeling and appearance development.

A. Studies on Pet-robot Interaction

Two relevant papers on owl-inspired pet robots and the related animal-robot interaction were available. The first paper presented the design, modeling and fuzzy control of a standing owl robot focused on mimicking head and neck rotations [21]. This study established a geometric and dynamic model for bionic head movement and verified the control performance via simulation, but the robot was designed for engineering verification rather than emotional companionship, lacked human–robot interaction, and had not been implemented with autonomous behavior or physical prototyping for daily companion use. The second study developed two types of owl-shaped pet robots for emotional communication and compared physical blinking and display-based eye expression [22]. This work was more directly applicable to our objectives, proposed an emotion generation model based on psychological space, and evaluated user recognition of emotional expressions, yet suffered from limited interaction modes, high complexity of hardwares, high power consumption, insufficient structural stability, and lack of a lightweight parallel control architecture.

The model in [23] focused on replicating the neural mechanism of barn owl auditory-visual localization and provided a biological basis for adaptive sensorimotor control, but it was restricted to orienting tasks and could not receive human touch, sound or complete intentional interaction. The research in [24] advanced biomimetic hardware by integrating neuromorphic circuits to speed up sensory adaptation, yet it retained the limitations of pure stimulus-driven response and did not offer enough variability in expressions or independent behaviors. A key difference between Animal-Robot Interaction for biological study and our companion robot lies in purpose: the former targets engineering verification of biological principles, while our owl-shaped pet companion robot centers on sustained emotional presence and natural human–robot bonding. Currently, no owl-inspired bionic robot provides low-burden, autonomous, multi-modal emotional companionship under cost and size constraints, highlighting the novelty of our cat-owl companion robot.

B. Owl Behavioral Studies

Extensive biological studies show that owls have fixed eyes[25] and rely on wide-range head rotations[26] to adjust their field of view, providing a simple and reliable motion mechanism for robotic design. Their wings move gently with a small amplitude of about 45° to 60°[27] during rest or attention, ensuring quiet and safe interaction. Owls are naturally calm, observant, and sensitive to surrounding sounds and movements, often keeping still for long periods while responding with subtle head tilts, mild vocalizations, and slight posture changes[28]. These stable, low-intensity, and highly anthropomorphic behaviors are highly suitable for emotional companion robots, enabling natural, non-repetitive, and comforting interactions without complex structures or exaggerated movements[29][30].

III. DESIGN REQUIREMENT

Given the intended use of the owl-shaped pet companion robot, the design could not be overly complex and could not require components that would be difficult to acquire for the average person. The robot design also needs adequate safety measures to prevent users from being injured during operations. Therefore, the robot is equipped with soft fur and without sharp edges.All components related to the control system, such as wires, needed to be specially designed, internally enclosed, and equipped with embedded interfaces to prevent users from directly accessing them.

IV. MATERIALS AND METHODS

A. Material List

The following materials were used to design, assemble, and test the owl robot system: ESP32 Dev Module controller (1), USB cable (1), computer with Arduino Integrated Development Environment(IDE)(1), SolidWorks Computer Aided Design (CAD) software, Open Source Fritzing Hardware Prototyping Software, thin-film pressure sensor (1), sound sensor (1), bottom button (1), SG92 servo motors (3), MAX98357A I2S amplifier module (1), speaker (1), breadboard (1), jumper wires, 5V battery/power supply, 3D-printed Polylactic Acid (PLA) body shell, 3D-printed PLA wings, 3D-printed PLA internal supports, PLA filament, rabbit fur outer covering, glue, screws, and basic assembly tools.

B. Methods

The following high-level steps were used:

- Step 1: 3D modelling and key parameters solutions
- Step 2: Basic control theorem and loop
- Step 3: Specific coding achievement
- Step 4: Possible bias and solutions while assembling
- Step 5: Performance Analysis

1) Designing the Robot:

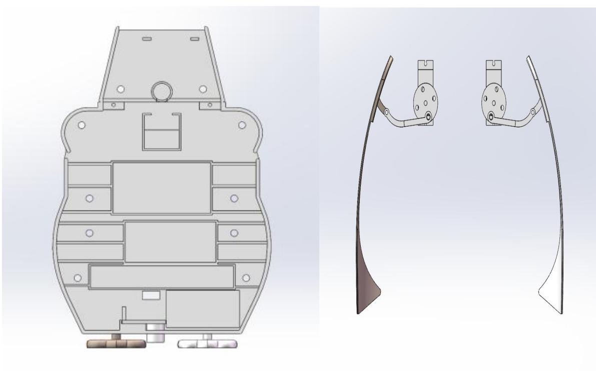
Following the initial conceptual design, the next phase focused on developing the robot's structural components and completing the assembly.

First, the robot's main body frame was designed as a single-piece, hollow shell with internal compartments to house electronic components, batteries, and wiring, as shown in the first image. The frame features a streamlined, animal-inspired silhouette, with mounting holes and recessed areas pre-cut for servomotors, connection brackets, and other functional parts to ensure a compact and tidy layout.

Next, the actuation system for the robot's side limbs was designed. As shown in the second image, each limb is driven by a servomotor connected to a linkage mechanism. The servomotor's rotational motion is transferred through a connecting rod to the limb panel, enabling controlled opening and closing movements.

The third image illustrates how the limb mechanisms are integrated with the main body frame. The servomotor mounts are fixed to the frame's side recesses, aligning the linkage system perfectly with the limb panels. This modular design ensures easy assembly, disassembly, and future maintenance.

Finally, the full external assembly of the robot was completed, as shown in the fourth image. The top head enclosure, main body shell, and side limb panels were combined to form a complete, smooth exterior. A wood-texture finish was applied in the CAD render to simulate the final appearance, showcasing the robot's rounded, friendly form factor.



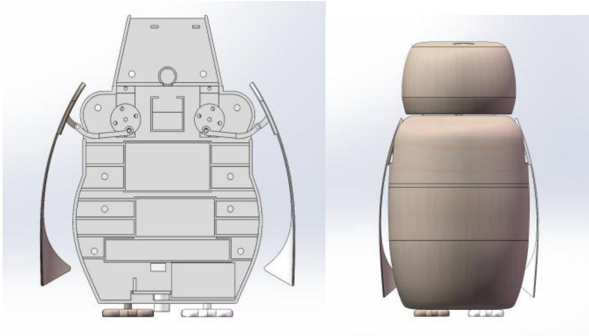


Fig. 2. 3D modelings in solidworks.

2)Building the Robot: During the design phase of the 3D-printed components, certain bolt holes were intentionally omitted from the 3D model to preserve adjustable tolerances. This approach enhances assembly flexibility and facilitates fine-tuning based on actual alignment conditions. The necessary bolt holes were measured and drilled individually during the assembly process.

To simplify the alignment of joint axes, the robot was assembled in an inside-out sequence, wherein the core structure was constructed first, followed by the gradual installation of peripheral components.

And The servo motor for the head was secured using a mortise-and-tenon joint in combination with screws. The servo motor responsible for driving the wing motion was fastened using a pen cap of appropriate width, together with a mortise-and-tenon joint and screws. Both servo motors were additionally reinforced with 502 cyanoacrylate adhesive. After the adhesive had fully cured, the wing structure and the head structure were respectively connected to their corresponding servo motors.

The final assembly step involved attaching the owl-shaped pet companion robot to a plywood base. To enhance its tactile and aesthetic qualities, the outer shell of the robot was covered with rabbit fur.

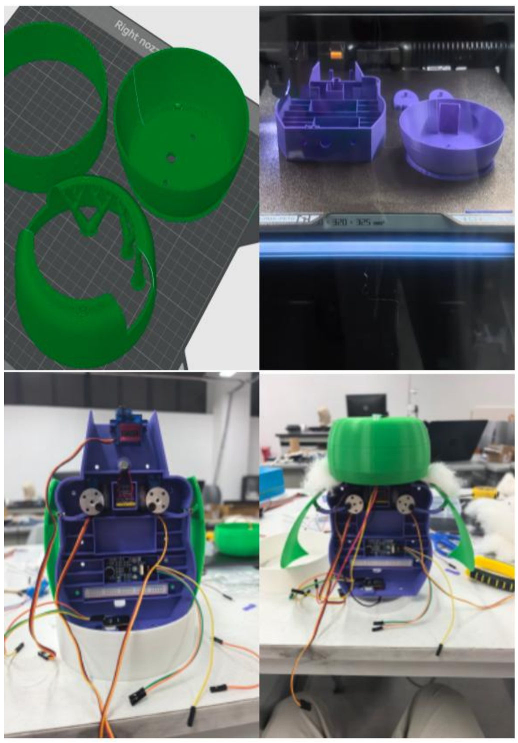


Fig. 3. Building process.

Regarding the electrical wiring and power supply, a portion of the breadboard was sectioned off, as this area proved sufficient for the circuit connections while simultaneously reducing the overall volume of the robot and minimizing material waste. A part of breadboard was used to establish connections among the Arduino board, , and the positive, negative, and control signal lines of all three servo motors. To ensure adequate and stable power for the servo motors, the breadboard was connected to an external 9V lithium battery. Figure 4 presents a wiring diagram generated using Fritzing, an open-source wiring illustration tool, to depict the connections required to power and control the core configuration of the owl-shaped robot.

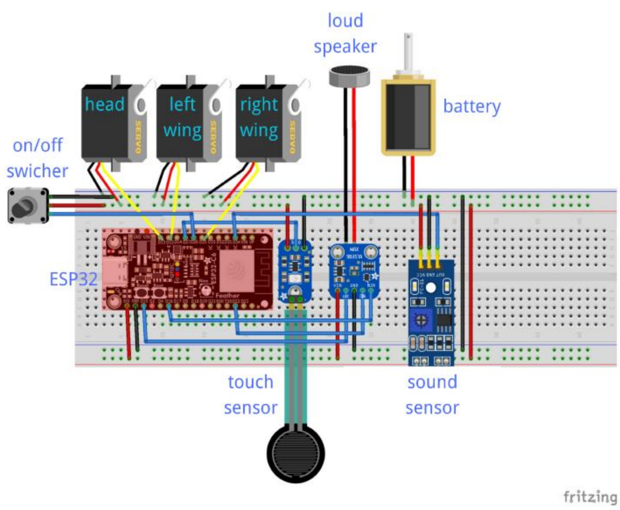


Fig. 4. wiring diagram.

*3)Estimate the height of the center of mass of the system:*The process for solving the kinematics is described in greater depth in Section V.

4) Kinematics of the wings: The process for solving the kinematics of the wings is described in Section VI.

5) Developing and Implementing Movement functions through the code: The process for is detailedly described in Section VII.

6) Experimentation and Testing (Evaluation): Initial testing of the automatic motion trigger timing on the fully assembled owl-shaped pet robot was generally successful, and the robot executed pre-programmed actions as expected. However, a significant mechanical discrepancy was observed during wing kinematics testing. During clockwise rotation, the right wing consistently exhibited an unintended counter-clockwise deviation of approximately 3 degrees per motion cycle; conversely, during counter-clockwise rotation, the left wing deviated an additional 3 degrees in the clockwise direction. As the number of motion cycles increased, the cumulative positional error between the two wings grew progressively, leading to failure in maintaining the robot's symmetrical flight posture.The table below records the deviation.

Cycles	**Left Wing Actual Angle (°)**	**Right Wing Actual Angle (°)**	**Angle Difference (°)**
5	31	28	+3
10	63	55	+8
15	96	81	+15

Table. 1. Wing flap angle deviation after multiple cycles.

The error margins for the remaining mechanical components were minimal and considered negligible, having no meaningful impact on overall functional assessment. The table below presents the measured automatic function(function3) motion trigger timing for the robot.

Item	Head Turn	Bird Chirp	Wing Flap
1st Trigger Interval	23 s	38 s	31 s
2nd Trigger Interval	47 s	12 s	19 s
3rd Trigger Interval	15 s	44 s	42 s
Average (s)	28.3	31.3	30.7
Standard Deviation (s)	13.6	13.7	9.5

Table. 2. The actual situation of automatic functions.

Thus, it can be seen that this function not only operates correctly—its execution logic has been verified and it is functionally reliable—but also exhibits sufficient stability in its external behavior, maintaining consistent actions and outputs across different environments and input conditions, thereby meeting our basic requirements for robustness and consistency.

7)Performance Analysis: We can basically consider this little owl to have met our requirements. It is preliminarily hypothesized that the primary cause of the following errors is the poor quality of the internal servo and linkage components, resulting in insufficient motion repeatability. Despite multiple calibration attempts at the software level, the cumulative tolerances from the hardware could not be fully compensated by algorithms. Due to practical constraints, we regret that we were unable to conduct complete clinical or standardized testing of this prototype in a professional laboratory setting. Nevertheless, all individuals who participated in the user experience, as well as visitors who observed the demonstration, provided positive feedback regarding the robot's overall design concept and dynamic performance, acknowledging its potential for further development and entertainment value. Testing of the robot was halted at this stage, with plans to reconstruct it using higher-precision servo systems in the future.

C. Asssumptions

The design and modeling assumptions are below:

• The encoders in the servomotors have sufficient precision for this task.

• All linkages were modeled as rigid linkages

• The transmission part of the wing is modeled as a passive joint that was frictionless, had no damping, and had no elasticity.

• For the robot, when solving the kinematics, both static and dynamic stability would be assumed at all points.

V. Estimate the Height of the Center of Mass of the System

A. Determine of the height of the center of mass(COM) of the body part through the suspension method

The fuselage of the owl robot is a uniform irregular rigid body. The suspension method is adopted to measure its COM height, which is based on the equilibrium principle of free suspension of rigid bodies: when a rigid body is freely suspended and remains static, its COM must lie on the vertical line passing through the suspension point.

It should be noted that the thickness of the fuselage has a negligible impact on the COM height, because its thickness direction is perpendicular to the vertical direction of the COM height and will not change the vertical coordinate of the COM.

To reduce experimental errors, three groups of parallel experiments are carried out, and the average is the final height. The calculation formula of the average value is as follows:

$$H_b = \frac{H_{b1} + H_{b2} + H_{b3}}{3}$$

B. Derive the formula for the center of mass height of the wing part

Each wing of the robot is composed of four components: support beam, wing surface, steering gear, and connecting disc. The wing's moving angle range is 0-45 degree, which affects the wing COM position.

The COM height of a single wing is:

$$H_w(\theta) = \frac{m_1 h_1(\theta) + m_2 h_2(\theta) + m_3 h_3(\theta) + m_4 h_4(\theta)}{m_1 + m_2 + m_3 + m_4}$$

in which

$$h_1(\theta) = \frac{y_A + y_B}{2}$$

$$h_2(\theta) = H_p - R_w \cos\theta$$

$$h_3(\theta) = H_s$$

$$h_4(\theta) = H_d$$

The robot has two symmetric wings, so the total COM height range of the two wings is consistent with that of a single wing.

Also, The wing thickness has a negligible impact on the robot's tipping stability, as it does not significantly change the vertical height of the overall COM or the horizontal projection position of the COM relative to the support base.

C. Estimate the variation range of the overall center of mass height

The overall COM height of the robot is the mass-weighted average of the fuselage COM height and the total COM height of the two wings, and the calculation formula is:

$$H_{\text{total}}(\theta) = \frac{M_b \cdot H_b + M_w \cdot H_w(\theta)}{M_b + M_w}$$

Substitute the known parameter values into the formula, and the variation range of the overall COM height of the robot can be obtained:

$$H_{\text{total}}(\theta) \in [69.4\,\text{mm}, 70.8\,\text{mm}]$$

According to the static stability margin formula

$$S = \frac{b}{2} - e$$

combined with the variation range of the overall COM height, the calculated static stability margin

$$S > 0$$

under all wing opening and closing angles. This indicates that the robot has sufficient static stability and no risk of collapse during static state and small-range movement.

VI. THE KINEMATICS OF THE WINGS

A. Mechanism Structure and Working Principle

The wings are the core design of this robot. We adopt a multi-stage transmission structure: the servo motor is connected to a turntable to first drive the movement of the inner wing joints, which in turn transmits power to and drives the main wing bodies. This structure enables more flexible wing movement and prevents internal parts from being exposed.

Nevertheless, this design makes wing motion hard to constrain, so we adopted double limiting measures. First, narrow gaps reserved on the body frame restrict the movement of inner joints. Second, externally attached fur connects the top of the wings to the head frame to guarantee an accurate spreading angle. Finally, we realized the bionic flapping motion of the wings under limited conditions.

The working principle is as follows: After the drive motor is started, its output shaft drives the eccentric crank to perform continuous uniform circular motion; the eccentric crank is connected to the rigid connecting rod through a hinge point, converting its own rotary motion into linear reciprocating push-pull motion of the connecting rod; the other end of the connecting rod is hinged to the wing rocker arm, thereby driving the wing rocker arm to perform fixed-axis reciprocating swing around the fixed root axis; the flexible outer skin wraps around the outside of the wing and acts as a geometric constraint structure. When the wing swings to the preset limit positions of 0° (fully folded) and 45° (fully extended), the structural tension and limiting effect of the outer skin will prevent the wing from rotating further, ensuring that the wing stably and repeatedly completes the opening and closing actions within the preset angle range, and avoiding component damage or motion failure caused by over-rotation of the mechanism.

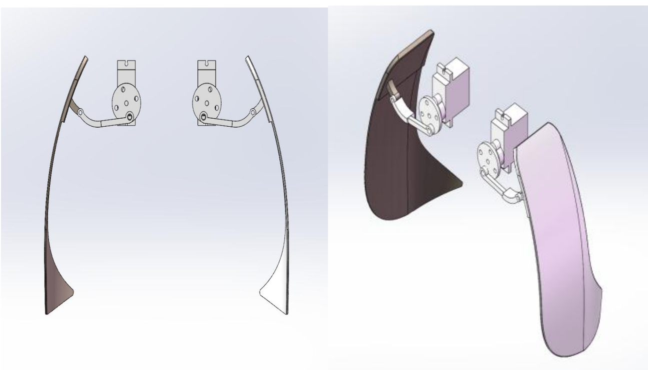


Fig. 5. The structure of the wing system.

B. Key Design Theoretical Basis

Since the center of mass of the wing does not coincide with the fixed root rotation axis, the moment of inertia of the wing about its own center of mass cannot be directly used for driving torque calculation. Therefore, the parallel-axis theorem is adopted to accurately calculate the moment of inertia of the wing about the root rotation axis, providing core basic parameters for subsequent motor torque matching and mechanism dynamic analysis.

The complete formula and derivation process are as follows:

$$I = I_{cm} + Md^2 = \rho \int_V r^2 dV + (\rho V)d^2$$

where r is the vertical distance from the differential mass dm on the wing to the centroid axis, and V is the overall volume of the wing. And if the wing is made of homogeneous material, then

$$dm = \rho dV$$

and

$$I_{cm} = \rho \int_V r^2 dV$$

According to the rigid body fixed-axis rotation law, combined with the dynamic characteristics of wing swing and considering the damping loss during rotation (damping coefficient is c), the improved torque balance formula and derivation process are as follows:

$$M = I\beta + c\omega = I\frac{d\omega}{dt} + c\omega$$

In the formula,

$$\sum M$$

is the total driving torque acting on the wing (the resultant force of the motor output torque and the frictional resistance torque), I is the total moment of inertia of the wing about the root rotation axis.And

$$\beta = \frac{d\omega}{dt}$$

is the angular acceleration , and $c\omega$ is the damping torque (used to correct the energy loss in actual motion).

At the same time, using the force arm amplification effect of the crank-connecting rod mechanism, the appropriate output torque of the DC motor is matched according to parameters such as crank eccentricity and connecting rod length, ensuring that the motor can transmit sufficient driving torque to the wing rocker arm through the crank-connecting rod transmission mechanism, realizing the wing to complete the preset opening and closing actions stably and quickly, and avoiding problems such as motion jamming and inability to reach the limit position caused by insufficient torque.

C. Motion Limit: Flexible Geometric Constraint

The limit positions of the wing's opening and closing are realized by the geometric constraint of the flexible outer fur. The flexible outer fur is made of a material with a certain toughness and tension, closely wrapping around the outside of the wing rocker arm, and its shape and size are designed according to the 0°~45° swing range of the wing. When the wing swings to 0° (fully folded), the outer skin is in a naturally relaxed state, limiting the wing from rotating further inward; when the wing swings to 45° (fully extended), the outer skin is stretched to the preset length, and its tension generates a reverse binding force, preventing the wing from rotating further outward, thereby realizing precise limiting of the wing swing, ensuring the consistency and repeatability of each opening and closing action, and improving the stability and reliability of the mechanism operation.

VII. Functions and code

A. Basic code and functions

The coding of the owl-shaped pet robot interaction system is a modular and event-driven program package developed based on C/C++ language, which is compatible with the Arduino/ESP32 development environment and serves as the core component for implementing the robot's control loop. This coding adopts a modular design, which is divided into functional files (OwlBot_Main.ino, config.h, events.h) and three core functional modules (sensors, scheduler, actuators).

1) The Role of the Coding: The core function of the code is to establish the connection between the hardware and software of the robot, converting all input signals into standardized events. It manages the priority of events, where user interaction events take precedence over autonomous behavior events, and adopts three states (IDLE, ACTIVE, COOLDOWN) for actuators to avoid action conflicts and mechanical overload. In addition, the code schedules autonomous events at random intervals of 10-50 seconds and supports modular debugging, which facilitates the maintenance of the system.

2)System Function of OwlBot Interaction System: The OwlBot interaction system has two core behavioral modes: human-triggered interaction and autonomous function. Human-triggered interaction refers to the robot's response to external inputs including touch, sound and button operations. Autonomous function means the robot performs random actions (such as head rotation, wing flapping and sound playback) at random intervals of 10-50 seconds without user input. Driven by the main loop, the system can stably handle the two behavioral modes, realizing the function of a lifelike companion robot.

B. Improving the control loop through coding

1)Initial Concept — Rule-based Behavior Design: The initial coding concept of the OwlBot interaction system was based on rule-based behavior design, with the core goal of realizing basic interaction and autonomous behavior functions. Specifically, the design included four key points: using sensors to detect user input signals, adopting ESP32 as the main control board to control servos and sound modules, setting state values to avoid repeated calls of actuators, and using random time intervals to realize autonomous behaviors of the robot. This initial design laid the foundation for the basic functions of the system but remained at a simple control level without considering potential conflicts and system stability issues.

2)Problem Discovery — Direct Control Was Not Enough: In the actual testing process of the initial rule-based coding, a series of defects of direct control were gradually exposed, which could not meet the requirements of stable and smooth operation of the system. The main problems included: multiple input signals might occur simultaneously, leading to confusion in system response; autonomous actions were likely to conflict with user touch operations, affecting user experience; sensors were prone to repeated triggering, resulting in invalid actions; the use of delay() function would block the entire program, reducing the real-time performance of the system; all codes were written in a single file, which was inconvenient for module debugging and problem location; servos and sound modules could not be tested independently, increasing the difficulty of troubleshooting.

3)System Redesign — Event-driven and Non-blocking Architecture: To solve the problems existing in direct control, the coding architecture was redesigned into an event-driven and non-blocking mode, with the core logic of "Sensors → Events → Dispatch → Actuator FSMs" and "Scheduler → Events → Dispatch → Actuator FSMs". Specifically, sensor inputs and autonomous timing signals generated by the scheduler were first converted into standardized events, which unified the input forms and avoided confusion caused by multiple input types. These events were then sent to the event dispatch mechanism, which judged whether to execute, delay or ignore the events according to the current state of the actuators and the preset event priority, effectively resolving conflicts between different events. At the same time, the non-blocking design based on millis() timing replaced the traditional delay() function, ensuring the real-time performance of system operation.

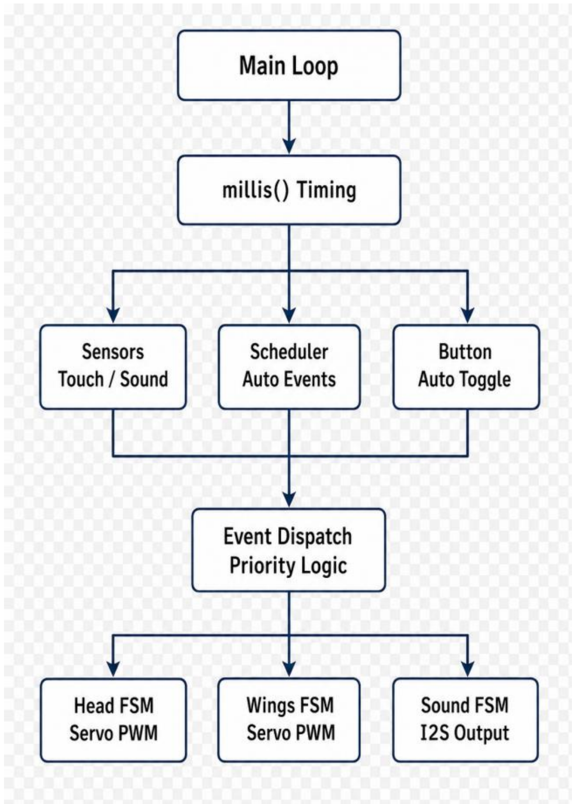


Fig. 5. System diagram.

4)Final Implementation — Modular Code and FSM-based Actuators: On the basis of the redesigned architecture, the final coding implementation focused on modular design and state machine-based actuator control. The code was split into multiple functional modules according to different functions, including the main program module (OwlBot_Main.ino), parameter configuration module (config.h), event definition module (events.h), sensor module, scheduler module and actuator module. Each module undertook independent functions, which realized the decoupling of code and facilitated independent debugging and maintenance of each part. For the actuator control, a finite state machine (FSM) design was adopted, defining three states (IDLE, ACTIVE, COOLDOWN) for each actuator, which effectively avoided repeated triggering of actuators and mechanical overload. Finally, the modular code and state machine-based actuator control realized the stable integration of user interaction and autonomous behavior, improving the reliability, maintainability and practicality of the system.

VIII. Conclusion and Future Development

A. Conclusions

We developed a simple owl-shaped pet companion robot to endow the people in need with warmth and companionship. Despite limited materials' quality, the concepts of the owl-shaped robot were shown to be viable. During operation, the robot can perform its functions normally and behave like a real owl. The study is new in the robotics field and possesses significant potential to be used enormously to enhance the entertainment, companionship, and well-being of companion animals[31].

B. Future Development

As expected, prototype models of the owl-shaped pet companion robot had many issues and often required multiple redesigns and repairs. Unfortunately, redesigns and more robust construction were beyond the scope of the research in its current iteration. The following issues were identified with the design and would need to be addressed in any future work.

The wings-flapping system is not natural enough and unstable, and a structural redesign of the robot with a more robust joint or additional support material is necessary for any further work. The new system in conjunction with improved modeling procedures would enable refinement of the control loop, and lead to better performance. Additionally, designing more complex movement functions would increase the level of being natural.

Due to the foot component, the robot needs to be placed on the ground and connected to the computer to get power. For a system that is expected to entertain pets, this is not acceptable. In future iterations, implementing control with a localized chip and batteries would be ideal. If this is achieved, the owl-shaped robot base could be set on a movable device for further autonomous function development.

In future versions, integrating voice interaction capabilities will be a key enhancement. The robot could be equipped with a microphone, along with a lightweight speech recognition module, allowing it to give non-verbal reponses. The current servo motors are also undersized for sustained flapping loads, leading to premature wear. Furthermore, incorporating flexible materials at the wing-root joints could better absorb impact forces and improve motion realism. Without these enhancements, the robot risks mechanical failure after only a few hours of continuous operation.

IX. Contribution Statement

It is presented as follows.

Name	Responsibilities	Contribution
Wang Zichen	3D Modeling, Initial control loop design	1/3
Zhang Ruihan	Main part of the coding, Developing control loop, PPT making	1/3
Huang Ziming	Actuator part of the coding, Testing, Report writing	1/3

Table. 3. Contribution Statement .

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